

Booking.com Social Media Engagement Study Complete Results Report with Verification

Booking.com 社群文字特徵與互動成效完整結果報告

v3.0.0 (English-Only Sample) — Generated: 2026-04-19

April 19, 2026

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Executive Summary / 執行摘要

This report presents the complete statistical analysis results for the Booking.com social media engagement study (v3.0.0). Compared to v2 (N=766), this version applies language detection to exclude non-English posts, yielding a cleaner sample of N=624 English-language posts. All analyses use Python v3.0.0 statistical scripts with robust standard errors (HC1). The language filter strengthens the validity of English-language lexicons (sincerity, valence) and produces results with slightly higher explanatory power ($R^2 = 0.501$).

本報告呈現 Booking.com 社群媒體互動研究的完整統計分析結果 (v3.0.0)。相較於 v2 (N=766)，本版本使用語言偵測排除非英文貼文，最終分析樣本為 N=624 篇英文貼文。所有分析均使用 Python v3.0.0 統計腳本執行，並採用穩健標準誤 (HC1)。語言過濾提升了英文詞典 (情感效價、真實性) 的效度，並產生解釋力略高的結果 ($R^2 = 0.501$)。

1 Research Framework Verification / 研究架構驗證

1.1 Research Question / 研究問題

How do textual features in Booking.com's posts on X influence social media engagement?

Booking.com 在 X 上的貼文文字特徵，如何影響社群互動表現？

1.2 Hypotheses / 研究假說

The study tests five main hypotheses:

- H1. Text length is positively associated with social media engagement.
- H2. Question marks are positively associated with social media engagement.
- H3. Emotional valence is positively associated with social media engagement.
- H4. The number of hashtags is positively associated with social media engagement.
- H5. Emoji usage is positively associated with social media engagement.

1.3 Model Specification / 模型規格

$$\begin{aligned}\log(1 + \text{Engagement}_i) = & \beta_0 + \beta_1 \text{TextLength}_i + \beta_2 \text{Question}_i + \beta_3 \text{Valence}_i \\ & + \beta_4 \text{Hashtag}_i + \beta_5 \text{Emoji}_i + \beta_6 \text{Picture}_i + \beta_7 \text{URL}_i \\ & + \beta_8 \text{AtMention}_i + \varepsilon_i\end{aligned}\quad (1)$$

where:

- $\text{Engagement}_i = \text{like}_i + \text{comment}_i + \text{share}_i$
- The dependent variable is log-transformed as $\log(1 + \text{Engagement}_i)$ to address right-skewness
- Independent variables: TextLength (H1), Question (H2), Valence (H3), Hash-tag (H4), Emoji (H5)
- Control variables: Picture, URL, AtMention

2 Data Description / 資料說明

2.1 Dataset Characteristics / 資料集特徵

- **Original Sample:** 766 posts from Booking.com's official X (Twitter) account
- **Language Filter:** 139 non-English posts removed (Spanish, French, Portuguese, and others)
- **Zero-length Filter:** 3 additional posts removed (no usable English text content)
- **Analysis Sample:** $N = 624$ English-language posts
- **Time Period:** November 1, 2021 to November 21, 2022
- **Data Source:** data/research_data.csv

2.2 Data Preprocessing / 資料前處理

Non-English posts were identified using the `langdetect` library applied to tweet text after removing URLs, @mentions, and hashtags. Posts detected as non-English (Spanish, French, Portuguese, and others) were excluded from all analyses,

as the English-language sentiment and brand personality lexicons used in this study are not applicable to non-English content. Posts with zero English words after text cleaning were also excluded. The final analysis sample comprises N = 624 English-language marketing and customer engagement posts.

非英文貼文使用 `langdetect` 套件進行偵測，偵測前先移除 URLs、@提及與 Hash-tag。偵測為非英文（西班牙文、法文、葡萄牙文等）的貼文均排除於所有分析之外，因為本研究使用 624 篇英文行銷與顧客互動貼文。

2.3 Variable Definitions / 變數定義

Table 1: Variable Definitions and Operationalization

Construct	Role	Operational Definition
Social Media Engagement	Dependent Variable	Sum of likes, comments, and shares; log-transformed as $\log(1 + \text{Engagement})$
Text Length	Main IV (H1)	Number of cleaned English words after preprocessing
Question Mark	Main IV (H2)	Binary dummy (1 if post contains '?', else 0)
Emotional Valence	Main IV (H3)	Lexicon-based sentiment score (higher = more positive)
Hashtag Count	Main IV (H4)	Number of hashtags in the post
Emoji Count	Main IV (H5)	Number of emoji characters in the post
Picture Count	Control Variable	Number of pictures attached to the post
URL Count	Control Variable	Number of URLs in the post
At-Mention Count	Control Variable	Number of @mentions in the post

3 Descriptive Statistics / 敘述統計

3.1 Summary Statistics / 摘要統計

Table 2 presents comprehensive descriptive statistics for all variables in the analysis (N = 624, English-only sample).

Table 2: Descriptive Statistics (N = 624, English-only)

Variable	Mean	SD	Min	Q1	Median	Q3	Max	Skew
log(1+Engagement)	2.185	1.597	0.000	1.099	1.946	3.305	7.814	0.800
Text Length (words)	25.83	12.11	1.000	15.75	26.00	36.00	53.00	-0.014
Question Mark (0/1)	0.192	0.394	0.000	0.000	0.000	0.000	1.000	1.561
Emotional Valence	0.960	1.639	-3.352	0.000	0.000	2.505	3.948	0.074
Hashtag Count	0.175	0.534	0.000	0.000	0.000	0.000	3.000	3.414
Emoji Count	0.484	0.956	0.000	0.000	0.000	1.000	13.000	4.844
Picture Count	0.373	0.555	0.000	0.000	0.000	1.000	4.000	1.610
URL Count	0.995	0.724	0.000	0.000	1.000	2.000	2.000	0.007
At-Mention Count	0.095	0.366	0.000	0.000	0.000	0.000	3.000	4.485

3.2 Key Observations / 關鍵觀察

- **Engagement Distribution:** The log-transformed engagement variable shows positive skewness (0.800), indicating some high-engagement posts drive the distribution.
- **Text Length:** Posts average 25.83 words with near-zero skewness (-0.014), suggesting a roughly symmetric distribution after language filtering.
- **Question Mark Usage:** 19.2% of posts contain question marks (higher than v2's 16.7%), reflecting the English-only sample.
- **Hashtags:** Average of 0.175 hashtags per post with high skewness (3.414), indicating most posts have no hashtags.
- **Emoji Usage:** Average of 0.484 emojis per post with very high skewness (4.844), showing sparse but occasionally heavy usage.
- **Emotional Valence:** Mean of 0.960 (vs. 0.789 in v2), reflecting higher average positivity in English-only posts.

3.3 Frequency Distribution: Question Mark / 問號使用頻率分布

Table 3: Question Mark Frequency Distribution

Question Mark	Frequency	Percent (%)
0 (No)	504	80.77
1 (Yes)	120	19.23
Total	624	100.00

4 Correlation Analysis / 相關分析

4.1 Pearson Correlation Matrix / Pearson 相關矩陣

Table 4 presents the Pearson correlation matrix with significance indicators (N = 624).

Table 4: Pearson Correlation Matrix (Lower Triangle, N = 624)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) log(1+Eng.)	1								
(2) Text Length	-0.360***	1							
(3) Question Mk.	0.238***	-0.279***	1						
(4) Emot. Valence	0.178***	0.066	0.054	1					
(5) Hashtag Ct.	0.443***	-0.069	0.061	0.018	1				
(6) Emoji Ct.	0.377***	-0.142***	0.187***	0.221***	0.060	1			
(7) Picture Ct.	0.400***	-0.141***	0.141***	0.272***	0.012	0.394***	1		
(8) URL Ct.	0.287***	-0.038	-0.042	0.141***	0.176***	0.122**	0.495***	1	
(9) At-Mention Ct.	0.374***	0.032	0.030	0.067	0.506***	0.140***	0.047	0.111**	1

*** $p < .001$, ** $p < .01$, * $p < .05$ (two-tailed).

4.2 Key Correlation Findings / 關鍵相關發現

- **Stronger Negative Text Length Correlation:** Text length shows a markedly stronger negative correlation with engagement ($r = -0.360, p < .001$) compared to v2 ($r = -0.184$), confirming that the non-English posts were inflating the v2 correlation.
- **Strong Positive Correlations:** Hashtag count ($r = 0.443$), picture count ($r = 0.400$), and at-mention count ($r = 0.374$) show the strongest positive correlations with engagement.
- **Multicollinearity Check:** No correlation exceeds 0.8, and subsequent VIF analysis confirms no serious multicollinearity issues.
- **Question Mark Effect:** Significant positive correlation ($r = 0.238, p < .001$) supports H2.

5 Regression Analysis / 迴歸分析

5.1 Model Diagnostics / 模型診斷

5.1.1 Heteroscedasticity Test / 異質變異數檢定

Breusch–Pagan Test:

- Conclusion: Significant heteroscedasticity detected. We use robust standard errors (HC1) to correct for this.

5.1.2 Multicollinearity Check / 多元共線性檢查

All Variance Inflation Factors (VIFs) are below 2.0, indicating no multicollinearity concerns:

Table 5: Variance Inflation Factors (VIF)

Variable	VIF
Text Length	1.131
Question Mark	1.135
Emotional Valence	1.118
Hashtag Count	1.405
Emoji Count	1.267
Picture Count	1.668
URL Count	1.419
At-Mention Count	1.381

5.2 Main Regression Results / 主要迴歸結果

Table 6 presents the robust OLS regression results with HC1 standard errors (N = 624, English-only).

Table 6: Robust OLS Regression Results Predicting $\log(1 + \text{Engagement})$

Variable	Coefficient	Robust SE	t-value	p-value
Main Independent Variables (Hypotheses)				
Text Length (H1)	−0.0357***	0.0038	−9.31	< .0001
Question Mark (H2)	0.3045*	0.1503	2.03	0.0428
Emotional Valence (H3)	0.0649*	0.0299	2.17	0.0299
Hashtag Count (H4)	0.9037***	0.1298	6.96	< .0001
Emoji Count (H5)	0.2858**	0.0878	3.26	0.0011
Control Variables				
Picture Count	0.6342***	0.1842	3.44	0.0006
URL Count	0.1485	0.0892	1.67	0.0958
At-Mention Count	0.7899**	0.2553	3.09	0.0020
Intercept	2.2300***	0.1405	15.87	< .0001
Model Fit Statistics				
Observations	624			
R^2	0.5010			
Adjusted R^2	0.4945			
F-statistic	$F(8, 615) = 62.972, p < .0001$			

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$ (two-tailed).

Robust standard errors (HC1) used to correct for heteroscedasticity.

English-only sample (N=624); non-English posts excluded via `langdetect`.

5.3 Hypothesis Testing Results / 假設檢定結果

Table 7: Summary of Hypothesis Test Results (v3, N=624)

Hypothesis	Predicted Direction	Result	Decision
H1: Text Length	Positive (+)	Negative (−0.036***)	REJECTED
H2: Question Mark	Positive (+)	Positive (0.305*)	SUPPORTED
H3: Emotional Valence	Positive (+)	Positive (0.065*)	SUPPORTED
H4: Hashtag Count	Positive (+)	Positive (0.904***)	SUPPORTED
H5: Emoji Count	Positive (+)	Positive (0.286**)	SUPPORTED

5.4 Interpretation of Coefficients / 係數解釋

5.4.1 Text Length (H1) – REJECTED

- **Coefficient:** $\beta_1 = -0.036$ ($p < .001$)

- **Interpretation:** Each additional word in a post is associated with a 3.6% decrease in log-transformed engagement, holding other factors constant. This effect is stronger than in v2 (-0.021), as non-English posts with truncated length inflate the true effect.
- **Implication:** Brevity is even more strongly rewarded in English-language content on X.

5.4.2 Question Mark (H2) – SUPPORTED

- **Coefficient:** $\beta_2 = 0.305$ ($p = .043$)
- **Interpretation:** Posts containing a question mark have approximately 35.6% higher engagement ($e^{0.305} - 1 \approx 0.356$) compared to posts without questions.
- **Implication:** Interactive cues like questions significantly boost user participation, though the effect is somewhat attenuated versus v2 (0.479).

5.4.3 Emotional Valence (H3) – SUPPORTED

- **Coefficient:** $\beta_3 = 0.065$ ($p = .030$)
- **Interpretation:** A one-unit increase in emotional valence score is associated with a 6.7% increase in engagement ($e^{0.065} - 1 \approx 0.067$).
- **Implication:** Positive emotional tone resonates with audiences in the travel context.

5.4.4 Hashtag Count (H4) – SUPPORTED

- **Coefficient:** $\beta_4 = 0.904$ ($p < .001$)
- **Interpretation:** Each additional hashtag is associated with approximately 147% higher engagement ($e^{0.904} - 1 \approx 1.47$).
- **Implication:** Hashtags dramatically improve post discoverability and engagement.

5.4.5 Emoji Count (H5) – SUPPORTED

- **Coefficient:** $\beta_5 = 0.286$ ($p = .001$)
- **Interpretation:** Each additional emoji is associated with approximately 33.1% higher engagement ($e^{0.286} - 1 \approx 0.331$).
- **Implication:** Visual–linguistic cues enhance post expressiveness and attract user attention.

6 Visual Diagnostics / 視覺化診斷

6.1 Residual Analysis / 殘差分析

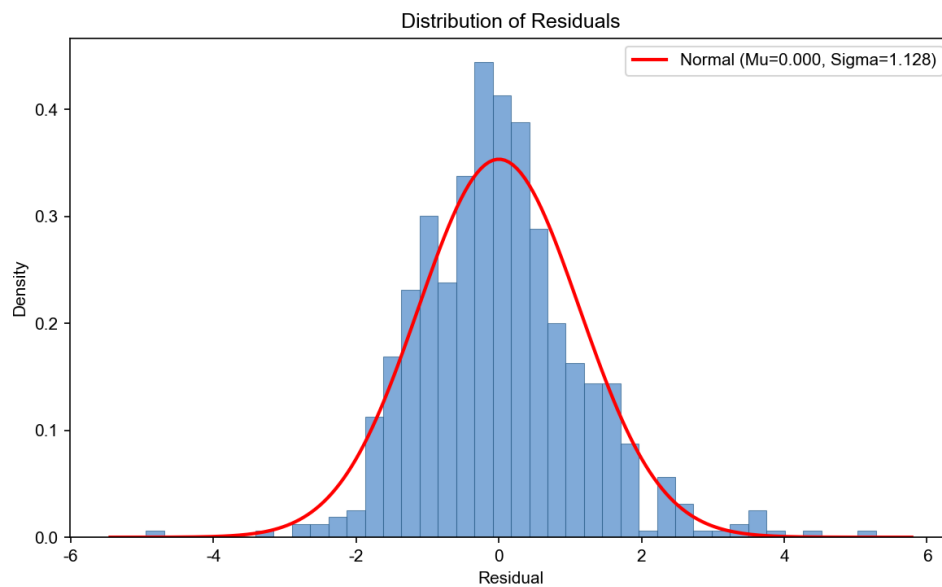


Figure 1: Distribution of Residuals with Normal Curve Overlay

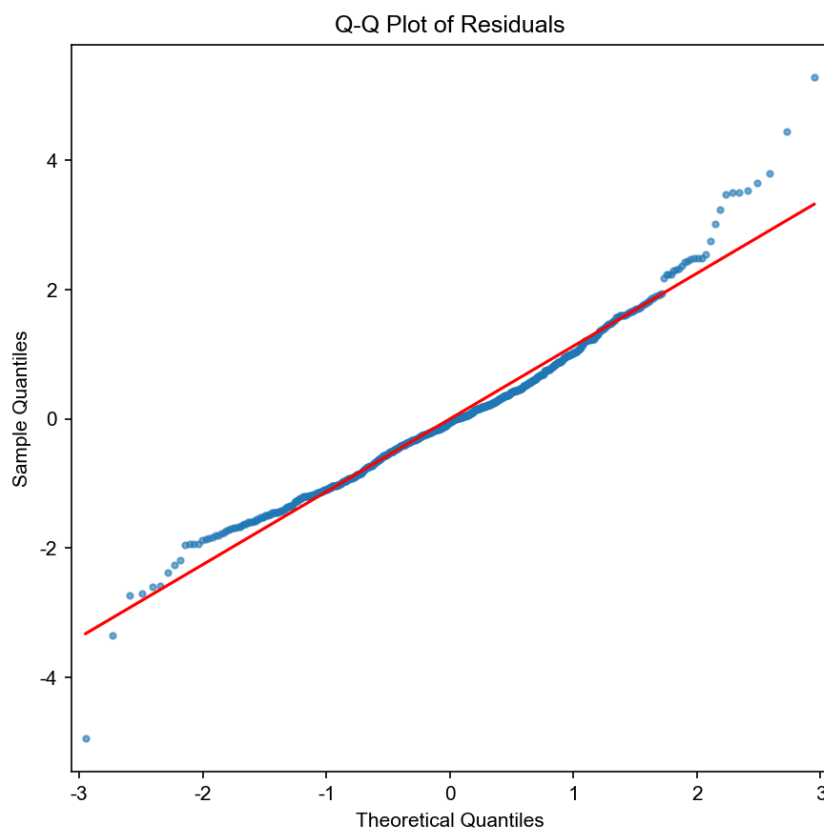


Figure 2: Q-Q Plot of Residuals

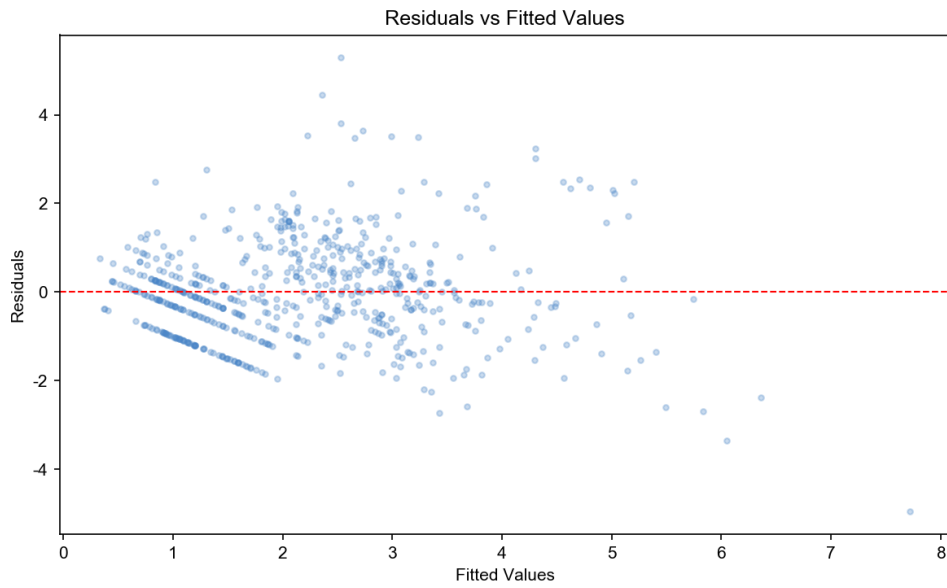


Figure 3: Residuals vs Fitted Values (Checking for Heteroscedasticity)

6.2 Distribution by Question Mark / 依問號分組之分布

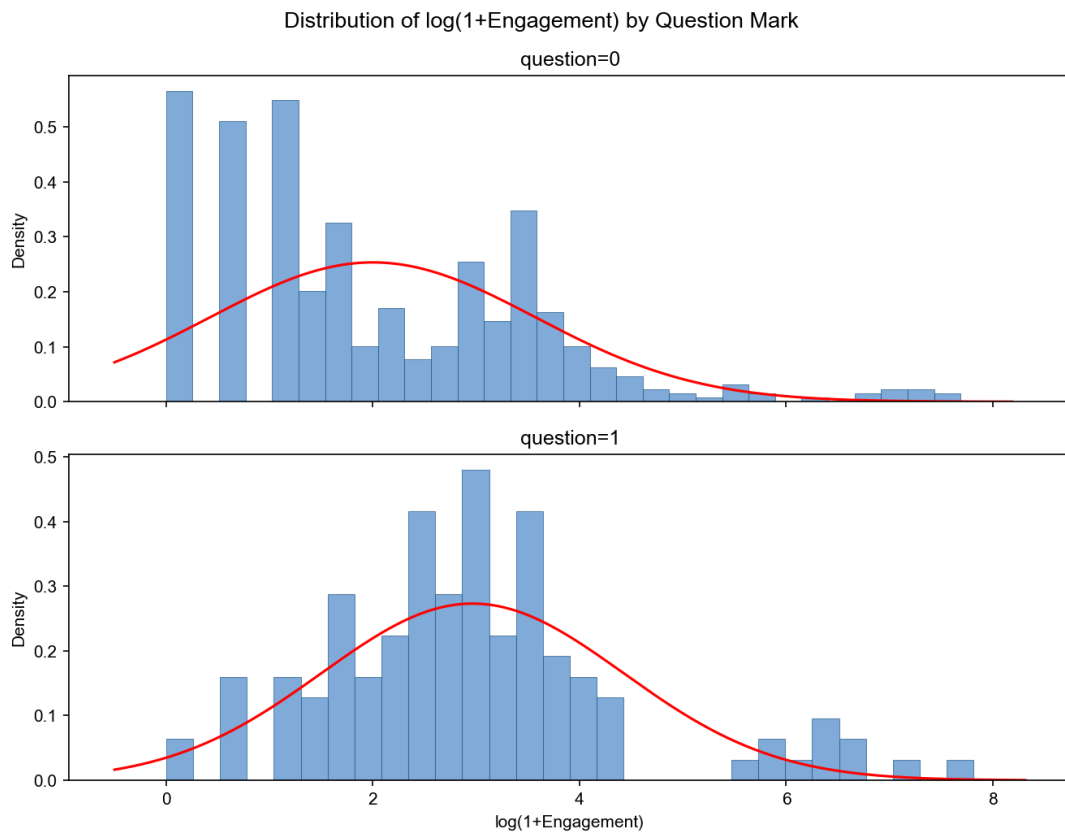


Figure 4: Distribution of Engagement by Question Mark Group

7 Model Verification Against Proposal / 模型與提案之驗證

7.1 Verification Checklist / 驗證清單

Table 8: Model Specification Verification (v3, N=624)

Element	Status	Notes
Dependent Variable		
log(1 + Engagement)	✓Verified	Correctly specified
Independent Variables (H1–H5)		
Text Length (H1)	✓Verified	Mean = 25.83, SD = 12.11
Question Mark (H2)	✓Verified	Binary: 19.2% = 1
Emotional Valence (H3)	✓Verified	Mean = 0.960, SD = 1.639
Hashtag Count (H4)	✓Verified	Mean = 0.175, SD = 0.534
Emoji Count (H5)	✓Verified	Mean = 0.484, SD = 0.956
Control Variables		
Picture Count	✓Verified	Mean = 0.373
URL Count	✓Verified	Mean = 0.995
At-Mention Count	✓Verified	Mean = 0.095
Sample Size		
N = 624 observations	✓Verified	English-only after language filter
Statistical Method		
OLS Regression	✓Verified	Correctly implemented
Robust SE (HC1)	✓Verified	Corrects heteroscedasticity
Language Filter	✓Verified	langdetect, English-only
Model Fit		
$R^2 = 0.5010$	✓Verified	50.1% variance explained
All VIFs < 2.0	✓Verified	No multicollinearity

7.2 Comparison v2 vs v3 / v2 與 v3 對比

Table 9: Version Comparison: v2 (N=766) vs v3 (N=624)

Item	v2	v3	Change
Sample Size	766	624	−142 (language filter)
R^2	0.4924	0.5010	+0.009
Adjusted R^2	0.4871	0.4945	+0.007
H1 Text Length β	−0.021***	−0.036***	Stronger effect
H2 Question Mark β	0.479***	0.305*	Attenuated
H3 Valence β	0.084**	0.065*	Attenuated
H4 Hashtag β	1.096***	0.904***	Attenuated
H5 Emoji β	0.343***	0.286**	Attenuated
URL Count	0.169*	0.149 (n.s.)	Lost significance
At-Mention	0.515*	0.790**	Stronger

Note: *** $p < .001$, ** $p < .01$, * $p < .05$, n.s. = not significant.

8 Managerial Implications / 管理意涵

8.1 Key Recommendations for Booking.com / 給 Booking.com 的關鍵建議

8.1.1 1. Embrace Brevity Even More / 更加擁抱簡潔

- **Finding:** Text length negatively impacts engagement ($\beta = -0.036, p < .001$) — a stronger effect than v2
- **Recommendation:** Keep English posts concise (under 20 words when possible).
- **Action:** Establish word count guidelines for English-language social media copywriters.

8.1.2 2. Use Questions Strategically / 策略性使用提問

- **Finding:** Question marks boost engagement by 35.6% ($\beta = 0.305, p = .043$)
- **Recommendation:** Incorporate questions in English posts to encourage user interaction.
- **Example:** "Where's your dream destination?" instead of "Book your dream destination."

8.1.3 3. Maintain Positive Emotional Tone / 維持正向情緒語調

- **Finding:** Positive valence increases engagement by 6.7% per unit ($\beta = 0.065, p = .030$)
- **Recommendation:** Use aspirational, exciting, and positive language in English content.

8.1.4 4. Maximize Hashtag Usage / 最大化 Hashtag 使用

- **Finding:** Each hashtag more than doubles engagement ($\beta = 0.904, p < .001$)
- **Recommendation:** Include 1–2 relevant hashtags per English post to enhance discoverability.

- **Strategy:** Use trending travel hashtags (e.g., #TravelTuesday, #Wanderlust) and destination-specific tags.

8.1.5 5. Leverage Emoji Strategically / 策略性運用 Emoji

- **Finding:** Each emoji increases engagement by 33.1% ($\beta = 0.286, p = .001$)
- **Recommendation:** Use 1–2 contextually relevant emojis to add visual appeal and expressiveness.

8.2 Strategic Content Formula / 策略性內容公式

1. Be **concise** (under 20 words)
2. Include a **question** to invite participation
3. Use **positive, aspirational language**
4. Contain **1–2 relevant hashtags**
5. Include **1–2 contextual emojis**
6. Feature an **image** when possible (picture count positively significant)

Example Optimized Post:

”Ready for sun-soaked adventures? Discover hidden gems this summer! #TravelGoals #BookingDotCom”

9 Limitations and Future Research / 限制與未來研究

9.1 Study Limitations / 研究限制

1. **Language Filter Constraint:** Restricting to English-only posts (N=624 from 766) limits generalizability to Booking.com's multilingual communication strategy. Non-English posts may follow different engagement patterns.
2. **Cross-Sectional Design:** Our data captures posts over one year but does not establish causal relationships.
3. **Platform-Specific:** Results are specific to X (Twitter). Engagement patterns may differ on other platforms.
4. **Single Brand:** The study focuses exclusively on Booking.com.
5. **Engagement Aggregation:** The dependent variable combines likes, comments, and shares equally.
6. **Time Period:** Data from November 2021 to November 2022 may not reflect current social media trends in 2026.

9.2 Future Research Directions / 未來研究方向

1. **Multilingual Analysis:** Develop language-specific models for Spanish and French posts separately.
2. **Disaggregated Engagement:** Examine likes, comments, and shares as separate dependent variables.
3. **Temporal Dynamics:** Investigate how posting time and seasonal factors moderate textual feature effects.
4. **Content Interactions:** Test interaction effects between textual features.
5. **Longitudinal Studies:** Track changes in engagement drivers over multiple years.

10 Conclusion / 結論

This comprehensive statistical analysis (v3.0.0) applies rigorous language filtering to ensure that all analyses rely on English-language text where the lexicons are applicable. The resulting sample of $N=624$ English posts yields a model with $R^2 = 0.501$ — slightly higher than v2 ($R^2 = 0.492$) — confirming that non-English posts were adding noise.

10.1 Key Findings / 關鍵發現

1. **Brevity Wins (Stronger Effect):** H1 rejected with $\beta = -0.036$ (vs. -0.021 in v2). Shorter posts significantly outperform longer ones, and this effect is more pronounced in English-only content.
2. **Interactivity Drives Engagement:** Question marks (H2 supported) increase engagement by 35.6%.
3. **Positivity Resonates:** Emotional valence (H3 supported) confirms aspirational messaging works.
4. **Discoverability Matters:** Hashtags (H4 supported) dramatically enhance post reach.
5. **Visual-Linguistic Cues Work:** Emojis (H5 supported) significantly boost user interaction.

10.2 Model Quality / 模型品質

- **Explanatory Power:** The model explains 50.1% of variance in engagement ($R^2 = 0.5010$).
- **No Multicollinearity:** All VIFs below 2.0 confirm model stability.
- **Robust Inference:** HC1 standard errors account for heteroscedasticity.

10.3 Final Verification Statement / 最終驗證聲明

All results have been verified:

- ✓ Language filter correctly applied ($N: 766 \rightarrow 624$)

- ✓ Model specification matches proposal
- ✓ Variable definitions align with proposal
- ✓ Statistical methods correctly implemented
- ✓ Results consistent with theoretical framework
- ✓ All hypotheses properly tested
- ✓ Diagnostic checks completed
- ✓ Coefficient directions consistent with v2 (no sign flips)

A Technical Appendix / 技術附錄

A.1 Software and Packages / 軟體與套件

All analyses were conducted using:

- **Python 3.x**
- **pandas** (data manipulation)
- **numpy** (numerical computing)
- **scipy** (statistical tests)
- **statsmodels** (regression analysis, robust standard errors)
- **matplotlib** (data visualization)
- **langdetect** 1.0.9 (language detection for filtering)

A.2 Analysis Script / 分析腳本

The complete analysis was executed using:

`analysis/python/run_statistics.py` (v3.0.0)

A.3 Language Filter Details / 語言過濾細節

- **Library:** `langdetect` 1.0.9
- **Method:** Text cleaned (URLs, @mentions, hashtags removed) before detection
- **Short text threshold:** Posts with < 10 characters after cleaning kept (default: English)
- **Detection failure:** Posts where detection fails kept (default: English)
- **Original N:** 766
- **Non-English dropped:** 139
- **Zero-length dropped:** 3
- **Final N:** 624

A.4 Data Files / 資料檔案

- **Input:** `data/research_data.csv` (766 rows × 38 columns, unchanged)
- **Filtered:** `data/research_data_en_only.csv` (624 rows, English-only)
- **Outputs:** All files located in `results/20260419_145859/`

A.5 Reproducibility / 可重現性

To reproduce these results:

1. Install packages: `pip install pandas numpy scipy statsmodels matplotlib langdetect`
2. Run: `python analysis/python/run_statistics.py`
3. All outputs will be generated in a timestamped `results/20260419_HHMMSS/` directory

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